

Single Mode Ensemble PIV: Theory and Reynolds Stress Estimation

1 Ensemble PIV Fundamentals

1.1 Sum of Correlation Approach

Traditional instantaneous PIV computes a cross-correlation for each image pair independently:

$$R_{AB}^{(i)}(\mathbf{r}) = A_i \star B_i \quad (1)$$

where A_i and B_i are the first and second exposure images for the i -th realization.

Ensemble PIV instead sums correlations over N image pairs:

$$R_{AB}(\mathbf{r}) = \frac{1}{N} \sum_{i=1}^N R_{AB}^{(i)}(\mathbf{r}) = \frac{1}{N} \sum_{i=1}^N A_i \star B_i \quad (2)$$

In Fourier space, this becomes:

$$\mathcal{F}(R_{AB}) = \frac{1}{N} \sum_{i=1}^N \mathcal{F}(B_i) \cdot \mathcal{F}^*(A_i) \quad (3)$$

1.2 Benefits of Ensemble Averaging

1. **Improved Signal-to-Noise:** Random noise cancels while coherent signal accumulates
2. **Smaller Interrogation Windows:** Fewer particles per window required when averaging over many realizations
3. **Enhanced Near-Wall Resolution:** Smaller windows enable measurements closer to boundaries
4. **Robustness:** Outliers and spurious peaks are suppressed by averaging

1.3 Trade-offs

The fundamental trade-off is the **loss of time development information**:

- Instantaneous PIV: N independent velocity fields, temporal dynamics preserved
- Ensemble PIV: One averaged correlation, only statistical quantities recovered

Ensemble PIV is therefore suited for **statistically stationary** flows where we seek converged mean and turbulence statistics rather than time-resolved dynamics.

2 Reynolds Stress Estimation from Correlation Peak Shape

2.1 Velocity Decomposition

Following Reynolds decomposition, instantaneous velocities are split into mean and fluctuating components:

$$U_t = \bar{U} + U'_t \quad (4)$$

$$V_t = \bar{V} + V'_t \quad (5)$$

where the overbar denotes ensemble (or time) averaging, and primes denote fluctuations.

2.2 Correlation Peak Broadening

In instantaneous PIV, all particles within an interrogation window move with approximately the same displacement, producing a sharp correlation peak. Over an ensemble of realizations with velocity fluctuations, particles experience a **distribution** of displacements centered on the mean.

The cross-correlation is the autocorrelation *convolved* with the velocity probability density function:

$$R_{AB} = R_{AA} \otimes \text{PDF}(\text{velocity}) \quad (6)$$

For turbulent flows with Gaussian velocity fluctuations:

$$R_{AB} = R_{AA} \otimes \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad (7)$$

where $\boldsymbol{\mu}$ is the mean displacement and $\boldsymbol{\Sigma}$ is the velocity covariance (Reynolds stress tensor).

2.3 Gaussian Variance Relations

The autocorrelation peak shape reflects only the particle image diameter:

$$\sigma_x^2 = d_x^2 \quad (8)$$

$$\sigma_y^2 = d_y^2 \quad (9)$$

The cross-correlation peak is broadened by velocity fluctuations:

$$\sigma_x^2 = d_x^2 + \overline{u'u'} \quad (10)$$

$$\sigma_y^2 = d_y^2 + \overline{v'v'} \quad (11)$$

$$\sigma_{xy} = \overline{u'v'} \quad (12)$$

By comparing autocorrelation and cross-correlation peak shapes, the Reynolds stress components can be extracted:

$$\overline{u'u'} = \sigma_{x,AB}^2 - \sigma_{x,AA}^2 \quad (13)$$

$$\overline{v'v'} = \sigma_{y,AB}^2 - \sigma_{y,AA}^2 \quad (14)$$

$$\overline{u'v'} = \sigma_{xy,AB} \quad (15)$$

3 Single Mode: Asymmetric Window Weighting

3.1 The Near-Wall Resolution Problem

Standard ensemble PIV uses symmetric windows: both Frame A and Frame B are extracted with identical interrogation regions. The measurement location corresponds to the window center, so the minimum distance to a wall is approximately half the window size. For a 64-pixel window, measurements cannot approach closer than ~ 32 pixels from the wall.

3.2 Single Mode Architecture

Single mode ensemble PIV uses **asymmetric weighting** between the two frames to achieve finer spatial resolution while maintaining FFT efficiency. Three window sizes are involved:

1. **Particle Window** (small, e.g., 16×16 or 8×8): The effective measurement region
2. **Sum Window** (large, e.g., 64×64): The FFT computation size
3. **Fitting Window** (optional, e.g., 32×32): Region extracted from correlation for Gaussian fitting

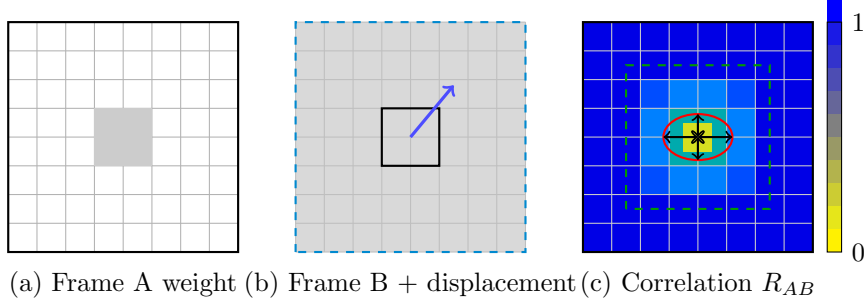


Figure 1: Single mode window weighting. (a) Frame A uses singlepix weighting: $W_A = 1$ only in the small central particle window (gray), $W_A = 0$ elsewhere. (b) Frame B uses uniform weighting over the full sum window (gray, cyan dashed outline); the arrow indicates particle displacement. (c) Resulting correlation plane with Gaussian peak; the green dashed line shows the fitting window region extracted for Gaussian fitting; the red ellipse and arrows indicate the fitted peak shape encoding displacement and Reynolds stresses.

3.3 The Singlepix Weighting Function

The key to single mode is the **singlepix** weighting applied to Frame A:

$$W_A(x, y) = \begin{cases} 1 & \text{if } (x, y) \in \text{particle window (centered)} \\ 0 & \text{otherwise} \end{cases} \quad (16)$$

Frame B uses uniform weighting over the full sum window:

$$W_B(x, y) = 1 \quad \forall (x, y) \in \text{sum window} \quad (17)$$

The weighted correlation becomes:

$$R_{AB}(\mathbf{r}) = \mathcal{F}^{-1} [\mathcal{F}(W_A \cdot A) \cdot \mathcal{F}^*(W_B \cdot B)] \quad (18)$$

3.4 Why Asymmetric Weighting Works

The particle window in Frame A defines *where* the measurement is localized—only particles within this small central region contribute to the correlation. The larger sum window in Frame B provides the *search range* for matching displaced particles.

This asymmetry allows:

- **Fine spatial resolution:** The measurement represents only the particle window region
- **Large displacement capture:** Particles can move anywhere within the sum window and still be matched
- **FFT efficiency:** Both frames are padded to the same sum window size for FFT computation

3.5 The Fitting Window (`sum_fitting_window`)

The full correlation plane from single mode has dimensions of the sum window (e.g., 64×64). However, Gaussian fitting only needs the central peak region. The **fitting window** extracts a smaller central portion (e.g., 32×32) for two reasons:

1. **Reduced computation:** Fewer points to fit
2. **Noise exclusion:** Correlation values far from the peak contain primarily noise

The fitting window must be large enough to capture the full Gaussian peak shape (typically $2\text{--}3\times$ the expected peak width).

3.6 Amplitude Normalization in Single Mode

Because Frame A uses a smaller effective area than Frame B, the cross-correlation amplitude is reduced by a factor of $\sqrt{\text{particle_area}/\text{sum_area}}$. When validating fitted parameters, this geometric factor must be accounted for:

$$R_{AB,\text{normalized}} = R_{AB} \times \sqrt{\frac{\text{sum_area}}{\text{particle_area}}} \quad (19)$$

4 The 13-Point Gaussian Fitting Procedure

4.1 Overview

To extract mean displacement and Reynolds stresses from the ensemble correlation, we fit 2D Gaussian functions to both the autocorrelation and cross-correlation peaks. A **13-point stencil** centered on the peak maximum provides robust parameter estimation with minimal computational cost.

4.2 Autocorrelation Fitting (R_{AA} , R_{BB})

The autocorrelation of each image with itself yields the particle image shape:

$$R_{AA}(\mathbf{r}) = A \exp \left(-\frac{(x - x_0)^2}{2\sigma_{x,A}^2} - \frac{(y - y_0)^2}{2\sigma_{y,A}^2} \right) \quad (20)$$

For symmetric particle images, $\sigma_{x,A} = \sigma_{y,A} = d_\tau$ where d_τ is the particle image diameter. The fit extracts:

- Peak location (x_0, y_0) — should be at origin for autocorrelation
- Widths $\sigma_{x,A}$, $\sigma_{y,A}$ — particle image diameter
- Amplitude A

4.3 Cross-Correlation Fitting (R_{AB})

The cross-correlation peak is fitted with a general 2D Gaussian including correlation:

$$R_{AB}(\mathbf{r}) = A \exp \left(-\frac{1}{2} \begin{bmatrix} x - \mu_x \\ y - \mu_y \end{bmatrix}^T \Sigma^{-1} \begin{bmatrix} x - \mu_x \\ y - \mu_y \end{bmatrix} \right) \quad (21)$$

where the covariance matrix is:

$$\Sigma = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{bmatrix} \quad (22)$$

The fit extracts:

- Peak location (μ_x, μ_y) — mean displacement
- Widths σ_x, σ_y — combined particle + velocity variance
- Covariance σ_{xy} — related to $\overline{u'v'}$
- Amplitude A

4.4 Joint Optimization Rationale

Rather than fitting autocorrelation and cross-correlation independently, a **joint optimization** approach offers advantages:

1. **Consistent Particle Diameter:** The same d_τ appears in both R_{AA} and R_{AB}
2. **Physical Constraints:** Reynolds stresses must be positive semidefinite
3. **Reduced Parameter Count:** Shared parameters reduce degrees of freedom
4. **Improved Robustness:** Joint fitting is less sensitive to noise in individual correlations

The joint objective function minimizes:

$$\mathcal{L} = \sum_{j=1}^{13} \left[(R_{AA,j} - G_{AA}(\mathbf{r}_j; d_\tau))^2 + (R_{AB,j} - G_{AB}(\mathbf{r}_j; \boldsymbol{\mu}, \boldsymbol{\Sigma}))^2 \right] \quad (23)$$

subject to the constraint that the velocity covariance $\boldsymbol{\Sigma} - d_\tau^2 \mathbf{I}$ is positive semidefinite.

5 Ensuring Physical Validity via Cholesky Decomposition

5.1 The Positive Semidefinite Requirement

A covariance matrix must be **positive semidefinite** (PSD) for physical validity:

$$\mathbf{x}^T \boldsymbol{\Sigma} \mathbf{x} \geq 0 \quad \forall \mathbf{x} \quad (24)$$

For the Reynolds stress tensor, this means:

- Diagonal elements (normal stresses) must be non-negative: $\overline{u'u'} \geq 0, \overline{v'v'} \geq 0$
- The determinant must be non-negative: $\overline{u'u'} \cdot \overline{v'v'} \geq (\overline{u'v'})^2$

Unconstrained fitting may produce unphysical results violating these conditions.

5.2 Cholesky Decomposition

Any positive semidefinite matrix $\boldsymbol{\Sigma}$ can be uniquely decomposed as:

$$\boldsymbol{\Sigma} = \mathbf{L} \mathbf{L}^T \quad (25)$$

where $\mathbf{L}$ is a lower triangular matrix with non-negative diagonal elements:

$$\mathbf{L} = \begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{bmatrix}, \quad L_{11} \geq 0, \quad L_{22} \geq 0 \quad (26)$$

This gives:

$$\boldsymbol{\Sigma} = \begin{bmatrix} L_{11}^2 & L_{11}L_{21} \\ L_{11}L_{21} & L_{21}^2 + L_{22}^2 \end{bmatrix} \quad (27)$$

5.3 Constrained Optimization via Cholesky Parameterization

Instead of optimizing Σ directly, we optimize the Cholesky factor $\mathbf{L}$:

1. Parameterize: $L_{11} \geq 0, L_{21} \in \mathbb{R}, L_{22} \geq 0$
2. Compute: $\Sigma = \mathbf{L}\mathbf{L}^T$
3. Evaluate: Objective function using this Σ

This **automatically guarantees** positive semidefiniteness without explicit inequality constraints. The optimization becomes unconstrained in L_{21} and simply bounded in L_{11}, L_{22} .

5.4 Recovering Reynolds Stresses

From the fitted Cholesky factors:

$$\overline{u'u'} = L_{11}^2 - d_\tau^2 \quad (28)$$

$$\overline{v'v'} = L_{21}^2 + L_{22}^2 - d_\tau^2 \quad (29)$$

$$\overline{u'v'} = L_{11}L_{21} \quad (30)$$

The particle diameter contribution d_τ^2 is subtracted since it appears in both auto- and cross-correlations.

6 Summary

1. **Ensemble PIV** sums correlations over many image pairs, trading temporal resolution for enhanced spatial resolution and statistical convergence.
2. **Correlation peak broadening** encodes velocity fluctuation statistics: the cross-correlation is the autocorrelation convolved with the velocity PDF.
3. **Reynolds stresses** are extracted by comparing autocorrelation and cross-correlation peak shapes:

$$\overline{u'u'} = \sigma_{x,AB}^2 - \sigma_{x,AA}^2, \quad \overline{v'v'} = \sigma_{y,AB}^2 - \sigma_{y,AA}^2, \quad \overline{u'v'} = \sigma_{xy,AB}$$

4. **Single mode** uses asymmetric window weighting—a small particle window for Frame A embedded in a larger sum window—to achieve fine spatial resolution while preserving FFT efficiency and displacement range.
5. **Gaussian fitting** extracts displacement and peak shape parameters; the optional fitting window reduces the correlation region used for fitting.
6. **Cholesky parameterization** of the covariance matrix guarantees physically valid (positive semidefinite) Reynolds stress estimates during optimization.